

No. of Functional Features Included in the Solution

Date	1 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Emotion Classification (BiLSTM)	Predicts learner emotion (Bored, Confident, Confused, Curious, Frustrated) from free text using a BiLSTM model	Model Engine (PyTorch)	Done	High
2	Emotion Classification (BERT)	Predicts learner emotion using a fine-tuned BERT transformer for comparison	Model Engine (Transformers)	Done	High
3	Mixed-Emotion Detection	Breaks down blended emotional states (e.g., Curious + Confused) instead of a single label	AI Engine	Done	Medium
4	Gemini AI Guidance Generator	Generates personalized tips, next steps, and encouragement based on detected emotion	Backend + Gemini API	Done	High
5	Model Comparison View	Displays BiLSTM vs BERT predictions side-by-side for transparency	Frontend UI (Streamlit)	Done	Medium
6	Interaction History Logging	Logs each interaction to CSV for continuous learning and record-keeping	Backend Storage	Done	Medium
7	Analytics Dashboard	Visualizes emotion trends over time for the learner	Frontend UI (Streamlit)	Done	Medium
8	AI Response Toggle	Lets users toggle the Gemini-generated response on or off	Frontend UI	Done	Low

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Model Comparison View, AI Response Toggle, Analytics Dashboard, Mixed-Emotion Display
2	Backend / Logic	2	Emotion Classification (BiLSTM), Emotion Classification (BERT)
3	Database / Storage	1	Interaction History Logging (CSV)
4	API / Integration	1	Gemini AI Guidance Generator
5	Security / Authentication	0	Not Used